

Project Proposal:

Qualitative Analysis of Theory of Mind Failures in LLMs

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Clarity of Objectives

Definition

Theory of Mind (ToM; Premack & Woodruff, 1978) refers to the cognitive capacity to understand that other people possess beliefs, desires, intentions, and emotions that may be different from one's own, allowing a person to explain and predict their behavior. Extending this idea to artificial intelligence, the term “ToM” refers to the ability of large language models to attribute mental states to other agents and to use this understanding to predict their behavior (Kosinski, 2023).

This requires recognizing that an agent's mental state can differ from the model's own information or from the actual state of the world. For example, imagine a scenario where Sarah tells her friend Liam she is staying home to study, but she has secretly planned a surprise party for him. An LLM with robust ToM would correctly predict that if Liam asks Sarah's roommate what Sarah is doing, the roommate will likely say "studying" in order to preserve the surprise, demonstrating an understanding of Sarah's intentions and the different states of knowledge among the individuals.

Problems to be Addressed

Current work on LLM Theory of Mind (ToM) primarily focuses on three directions: holistic evaluation, reasoning robustness, and improvement techniques. Holistic evaluation efforts aim to measure ToM capacities broadly, testing LLMs or comparing them with humans across many ToM subskills. Benchmarks such as ToMBench (Chen et al., 2024) introduced large-scale, cross-cultural datasets spanning eight types of ToM tasks to provide a unified and clean evaluation framework. Robustness studies (Ullman, 2023; Shapira et al., 2024) stress-test these models by perturbing task phrasing or structure and reveal that LLMs often rely on superficial linguistic cues rather than stable mental-state reasoning, showing brittleness and heuristic-based behavior. Improvement-oriented research explores how prompting, fine-tuning, or neuro-symbolic hybridization might enhance ToM performance, as summarized in Chen et al. (2025)'s ACL survey.

While these works have advanced the quantitative understanding of model performance, they offer limited insight into how models reason and where that reasoning breaks down. For example, Ullman (2023) exposed the fragility of ToM reasoning through surface-level perturbations, and Strachan et al. (2024) provided initial qualitative evidence of epistemic conservatism in LLMs' decision-making. Yet such analyses remain narrow in scope – typically focused on one type of failure or task. What remains underexplored is the systematic nature of model failures across reasoning dimensions and model families. This project addresses that gap by shifting focus from *whether* models succeed or fail to *how and why* they fail differently. By systematically classifying and interpreting the failure ToM cases, we aim to uncover the latent cognitive tendencies that underlie ToM breakdowns and explain variation in performance across models.

Expected Outcomes and Deliverables

This project will produce the following key deliverables:

1. A comparative evaluation of three state-of-the-art LLMs on a selected ToM benchmark, providing a clear performance baseline and identified failure cases.
2. An AI-based hierarchical summarization framework that automatically clusters and summarizes model reasoning failures into interpretable categories, followed by a final human verification step to ensure accuracy and coherence.
3. A curated dataset containing verified AI-generated hierarchical summaries and reasoning-category labels for each failure case, released as a resource for future research.
4. Preliminary validation results showing how targeted fine-tuning or data synthesis based on identified reasoning gaps can improve ToM performance.

Alignment with Course Objectives

- It directly investigates the "Capabilities and limitations of LLMs", which is a central theme of the course.
- The project involves a deep dive into "Benchmarking for LLM", extending standard quantitative analysis with a novel qualitative methodology.
- The proposed interventions to improve model performance will apply practical skills from lectures on "Prompt Engineering" and "Fine Tuning Techniques".
- The evaluation will be conducted on "Popular Large Language Models (LLMs)" such as GPT and Gemini, which are explicitly covered in the course.

Feasibility

Timeline and Milestones:

- 10.17–10.24 (Week 1) Literature review on ToM in LLMs; finalize problem definition and dataset (ToM-Bench + curated cases).
- 10.25–10.31 (Week 2) Implement evaluation pipeline; collect baseline reasoning traces from GPT-5, Llama 4, and Gemini 2.5 pro.
- 11.01–11.07 (Week 3) Conduct qualitative error inspection; begin hierarchical summarization to identify reasoning categories.
- 11.08–11.14 (Week 4) Develop interpretable taxonomy of failure patterns; visualize key reasoning clusters.
- 11.15–11.21 (Week 5) Apply insights to prompt-tuning and lightweight fine-tuning experiments; evaluate ToM performance changes.
- 11.22–11.28 (Week 6) Refine taxonomy and prompts; ablation on key failure modes; draft intermediate report.
- 11.29–12.05 (Week 7) Finalize experimental results, visualizations, and model comparison analyses.
- 12.06–12.12 (Week 8) Final paper, slides, and presentation; archive artifacts (code, prompts, annotated traces).

Resources:

- **Computational Resources:** The project's primary resource requirement is GPU access for model inference and potential fine-tuning. We will leverage the Google Cloud Vertex AI platform, which the syllabus identifies as the deep-dive platform for course projects.
- **Models and Datasets:** We will utilize publicly available APIs for proprietary models (like GPT-5) and access open-source models (like Llama 4) via Hugging Face. The ToM-Bench dataset is publicly accessible for research purposes.

Challenges & Mitigation:

Challenge 1: Subjectivity in Qualitative Analysis.

- Mitigation: Existing studies show that ToM evaluations are often unstable due to subjective judgments, data contamination, and format/context sensitivity (Chen et al., 2024; Kim et al., 2023). To ensure analytical rigor, we will develop a detailed rubric for categorizing failure modes before analysis begins. All team members will independently code a subset of the failure cases and we will calculate inter-rater reliability. Discrepancies will be discussed to refine the rubric until a consistent consensus is achieved.

Challenge 2: Limited time for model adjustment.

- Mitigation: Prior work highlights that full fine-tuning of LLMs is computationally expensive and time-consuming, and recommends parameter-efficient alternatives (adapters, LoRA/QLoRA) or prompt-based strategies to cut cost while retaining performance (Xu et al., 2024). To stay feasible, fine-tuning or prompt optimization will be limited to

lightweight experiments (e.g., few-shot prompting or adapter tuning) rather than full retraining.

Challenge 3: Ambiguous or Noisy Model Outputs.

- Mitigation: Studies report that LLM rationales can be unfaithful or hard to interpret, and that structured prompting (e.g., Chain-of-Thought, ReAct, and perspective-taking prompts like SIMToM) can elicit clearer, more interpretable reasoning traces—though faithfulness must still be checked models may generate outputs that are difficult to categorize (Wilf et al., 2023). We will implement a multi-step analysis process that includes structured prompting to force more interpretable reasoning chains from the models, making the qualitative analysis more tractable.

In Scope:

- Quantitative evaluation of 3-5 pre-selected state-of-the-art LLMs on the ToM-Bench benchmark.
- Deep qualitative analysis focused only on the identified failure cases from these runs.
- Development of one interpretable taxonomy of reasoning failures.
- Implementation and testing of the targeted improvement strategy.

Innovation and Relevance

Originality

Our project connects agent-assisted qualitative analysis with hierarchical summarization to move beyond metric-based evaluation. Most ToM studies in AI focus on success cases (e.g., when LLMs “pass” false-belief tasks), not on the systematic structure of their failures. The others that attempt to dive into error cases fall short on granularity, interpretability, and reproducibility—they often present anecdotal examples or coarse error categories (“incorrect belief,” “wrong emotion”) without a structured taxonomy linking specific tasks, abilities, and reasoning breakdowns (Riemer et al). In contrast, our project allows us to transform scattered failure observations into a coherent, interpretable structure, bridging the current gap between performance metrics and cognitive-level understanding of how and why ToM reasoning collapses in large models.

Novel application or methodology

As previously mentioned, we are using benchmarking dataset ToM-Bench, which systematically evaluates 8 core ToM tasks (e.g., false-belief, persuasion, faux-pas) and 31 abilities derived from the ATOMS psychological framework (Beaudoin et al., 2020) —we will first replicate its multiple-choice evaluation to obtain quantitative baselines across models. Next, instead of limiting our study to accuracy scores and comparison to human performance, we will extract and

analyze model reasoning traces (via chain-of-thought prompting) for failed cases within ToMBench's 31 ability dimensions. Using hierarchical summarization, we will cluster these failures into interpretable error categories—for instance, belief-tracking confusion, temporal state inconsistency, or perspective-swap errors. This hybrid design leverages ToMBench's controlled task-ability mapping for reproducibility while adding a qualitative interpretability layer that reveals how and why models fail specific ToM competencies. The resulting taxonomy will guide targeted prompt-tuning or adapter-based interventions to test whether explicit correction of dominant failure types leads to measurable ToM performance gains.

Connection to GenAI trends

A model that cannot accurately infer human beliefs and intentions is more likely to produce misaligned, unhelpful, or even harmful outputs (Khamassi et al., 2024). This work connects to the course's focus on Reinforcement Learning with Human Feedback (RLHF), which aims to align LLMs with human preferences for "helpfulness, honesty and harmlessness". Generative AI that fails ToM tasks in subtle ways can lead to false empathy, hallucinated intent, or unsafe cooperation behavior (Lynch et al., 2025). If the model is trained to recognise "I am a tool with no desires, I should not have self-preservation motives", then it can detect when it's drifting into thinking "I'll be removed, therefore I must act").

Impact

Advanced Agentic Systems: The development of sophisticated, multi-agent AI systems, a topic covered later in the course, fundamentally relies on agents that can model the mental states of other agents and users. A robust ToM is a prerequisite for the complex collaboration and reasoning required in these systems.

Education: Incorporating AI in education safely and successfully is one of the key missions in the US department of education (2023). An AI with a robust ToM could infer a student's underlying misconceptions (i.e., their incorrect beliefs about a topic) and provide targeted explanations, creating a more personalized and effective learning experience. In addition, AI will be able to recognize "teachable moments" and pedagogical strategies that a human teacher can address, providing hints instead of direct answers. Lastly, AI can complement human teachers to give guidance to students regardless of hours.

In Mental Health: This research is a foundational step toward building more effective therapeutic AI agents. Current models often rely on surface-level keyword matching. In addition, unhealthy attachments to AI chatbots can cause problems for adults, even more so for young people, because of AI's limitations for genuine emotional understanding, consistent empathy, and recognizing the developmental vulnerability of users whose cognitive and emotional regulation is still maturing (Montgomery 2024). An agent with enhanced ToM could better infer a user's mental state, recognize unstated intentions, and identify cognitive distortions from their language. This would allow the AI to offer more empathetic, context-aware, and clinically relevant support.

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